a

BALL — foundation-model performance

Classifiers

	Accuracy					Macro-F1				
	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM
DE baseline										
DE	97.6 ±0.5	N/A	N/A	N/A	N/A	95.0 ±1.2	N/A	N/A	N/A	N/A
BulkFormer										
Full	70.9	82.8	78.0	74.2	59.5	47.2	64.9	56.8	60.4	33.3
PCA	77.6	82.2	79.9	79.2	65.8	60.9	66.6	63.8	66.6	53.4
Supervised	68.8	81.3	76.9	72.3	65.6	42.9	62.1	59.7	55.5	42.7
Supervised + PCA	78.6	79.2	78.0	79.0	66.7	63.9	65.4	61.6	61.3	53.9
Full tokens	N/A	N/A	92.9	N/A	N/A	N/A	N/A	84.4	N/A	N/A
CellFM 1-view										
Full	14.3	43.6	22.2	19.3	13.0	4.5	29.2	13.7	11.7	3.5
PCA	18.4	26.6	26.8	25.4	18.9	7.3	17.1	14.5	13.9	6.4
Supervised	10.9	24.5	21.6	22.0	15.9	2.9	17.1	13.4	12.5	4.6
Supervised + PCA	18.2	16.4	19.7	18.2	17.6	9.6	10.6	10.8	10.9	6.2
Full tokens	N/A	N/A	14.9	N/A	N/A	N/A	N/A	5.8	N/A	N/A
CellFM 20-view										
Full	88.1	96.0	81.6	58.9	86.6	77.7	93.1	61.3	44.0	71.8
PCA	93.5	95.0	94.3	94.3	93.1	88.0	91.3	88.0	88.7	88.5
Supervised	84.1	86.0	83.4	80.3	82.4	72.7	72.8	75.0	66.6	72.0
Supervised + PCA	87.0	88.7	87.4	87.0	84.7	77.4	78.5	76.6	75.5	74.8
Full tokens	N/A	N/A	84.7	N/A	N/A	N/A	N/A	62.7	N/A	N/A
BulkRNABert										
Full	53.9	76.1	63.5	50.3	37.5	40.2	61.0	42.4	38.9	22.3
PCA	62.5	71.7	64.2	65.2	60.4	46.4	54.6	50.9	49.6	44.5
Supervised	54.3	73.8	63.3	51.2	41.5	35.8	59.5	44.8	40.0	25.3
Supervised + PCA	60.4	70.9	61.0	62.7	61.2	45.9	54.1	45.6	50.5	47.1

b

BRCA-TCGA — foundation-model performance

Classifiers

	Accuracy					Macro-F1				
	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM
DE baseline										
DE	86.4 ±1.2	N/A	N/A	N/A	N/A	83.3 ±1.1	N/A	N/A	N/A	N/A
DE · corrected	88.5 ±1.7	N/A	N/A	N/A	N/A	85.4 ±2.2	N/A	N/A	N/A	N/A
BulkFormer										
Full	84.9	84.3	83.6	84.9	78.6	82.0	81.2	80.1	79.6	73.5
Full · corrected	76.4	80.6	81.2	84.7	79.2	72.8	77.8	80.1	83.7	76.8
PCA	84.3	86.2	85.5	83.0	83.0	80.8	83.6	83.9	80.2	77.9
PCA · corrected	74.3	81.9	81.2	82.6	79.9	70.4	78.9	78.4	81.3	76.2
Supervised	83.6	81.1	83.6	81.1	80.5	75.3	74.9	81.2	76.5	77.2
Supervised · corrected	73.6	81.2	80.6	81.2	80.6	68.1	78.8	77.6	78.1	76.5
Supervised + PCA	79.9	82.4	82.4	81.1	79.9	74.7	78.1	78.1	74.9	75.1
Supervised + PCA · corrected	72.9	79.9	76.4	76.4	79.9	64.1	75.4	72.2	72.5	74.7
Full tokens	N/A	N/A	84.3	N/A	N/A	N/A	N/A	83.5	N/A	N/A
Full tokens · corrected	N/A	N/A	79.2	N/A	N/A	N/A	N/A	78.4	N/A	N/A
CellFM 1-view										
Full	53.5	59.7	56.6	54.7	50.9	41.5	47.2	41.5	42.2	16.9
Full · corrected	48.6	51.4	52.8	56.9	50.7	28.0	37.3	36.6	41.7	16.8
PCA	55.3	48.4	56.0	52.8	58.5	34.4	41.6	45.8	46.2	33.0
PCA · corrected	59.0	51.4	54.9	57.6	57.6	35.2	44.4	44.4	46.5	34.6
Supervised	50.3	42.1	49.1	45.9	54.1	20.7	33.2	35.8	32.7	29.4
Supervised · corrected	47.2	41.0	43.8	41.7	47.2	34.8	35.4	33.9	31.3	22.6
Supervised + PCA	49.7	48.4	49.1	49.7	56.6	25.5	38.0	39.8	38.8	39.5
Supervised + PCA · corrected	52.8	49.3	41.0	41.0	45.1	22.3	41.7	31.5	30.8	27.6
Full tokens	N/A	N/A	45.3	N/A	N/A	N/A	N/A	24.2	N/A	N/A
Full tokens · corrected	N/A	N/A	41.7	N/A	N/A	N/A	N/A	20.6	N/A	N/A
CellFM 20-view										
Full	81.8	82.4	74.8	76.1	78.0	79.5	79.5	55.4	71.4	65.7
Full · corrected	80.6	84.0	77.1	77.8	77.8	76.0	79.9	68.4	69.1	65.5
PCA	80.5	80.5	83.0	83.6	81.8	77.9	78.0	80.2	80.8	78.9
PCA · corrected	84.7	81.9	81.2	81.9	77.8	80.8	78.9	78.9	78.9	72.5
Supervised	82.4	78.0	76.7	75.5	79.2	79.1	76.5	71.4	67.3	73.3
Supervised · corrected	78.5	79.9	77.1	69.4	81.2	72.5	75.5	68.5	63.8	73.9
Supervised + PCA	79.2	74.8	76.1	77.4	78.0	75.9	70.4	73.4	75.3	72.4
Supervised + PCA · corrected	79.2	74.3	79.2	76.4	76.4	73.0	73.7	75.3	73.6	71.0
Full tokens	N/A	N/A	67.9	N/A	N/A	N/A	N/A	44.8	N/A	N/A
Full tokens · corrected	N/A	N/A	70.8	N/A	N/A	N/A	N/A	50.1	N/A	N/A
BulkRNABert										
Full	76.7	79.2	78.6	76.1	78.0	66.8	69.5	70.8	69.7	69.5
Full · corrected	75.7	77.8	77.1	74.3	75.0	70.0	72.4	70.2	67.1	67.3
PCA	79.2	80.5	81.8	83.0	81.8	68.9	74.1	74.9	76.5	76.2
PCA · corrected	70.8	76.4	79.9	76.4	75.7	62.1	70.3	72.7	72.3	63.7
Supervised	77.4	79.9	79.9	76.7	78.0	68.5	70.1	68.3	70.3	68.6
Supervised · corrected	75.0	76.4	76.4	74.3	74.3	67.5	70.9	69.9	67.9	65.1
Supervised + PCA	80.5	79.2	81.8	78.0	79.9	70.9	73.8	74.5	71.3	70.3
Supervised + PCA · corrected	75.7	73.6	78.5	78.5	75.0	71.4	69.0	71.3	73.9	62.6

c

BRCA-TCGA-APOLLO — foundation-model performance

Classifiers

	Accuracy					Macro-F1				
	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM
DE baseline										
DE	74.1 ±4.2	N/A	N/A	N/A	N/A	78.0 ±3.9	N/A	N/A	N/A	N/A
DE · corrected	87.6 ±4.0	N/A	N/A	N/A	N/A	84.9 ±5.6	N/A	N/A	N/A	N/A
BulkFormer										
Full	75.0	75.0	79.4	72.1	79.4	54.8	70.4	76.5	66.4	57.4
Full · corrected	85.3	86.8	85.3	80.9	83.8	80.6	80.9	77.9	67.7	78.2
PCA	61.8	72.1	75.0	69.1	60.3	38.5	68.5	58.5	56.8	40.6
PCA · corrected	80.9	85.3	86.8	86.8	83.8	73.1	79.6	84.0	83.1	77.7
Supervised	76.5	72.1	82.4	82.4	77.9	53.0	66.6	77.9	68.7	59.9
Supervised · corrected	82.4	79.4	82.4	79.4	89.7	77.2	72.0	75.7	66.8	87.4
Supervised + PCA	69.1	73.5	64.7	66.2	60.3	58.7	66.9	40.0	36.8	37.7
Supervised + PCA · corrected	80.9	85.3	80.9	80.9	77.9	73.0	79.9	74.2	73.2	68.7
Full tokens	N/A	N/A	80.9	N/A	N/A	N/A	N/A	84.4	N/A	N/A
Full tokens · corrected	N/A	N/A	82.4	N/A	N/A	N/A	N/A	78.2	N/A	N/A
CellFM 1-view										
Full	57.4	54.4	51.5	54.4	52.9	40.0	38.2	34.4	36.7	17.3
Full · corrected	45.6	57.4	58.8	55.9	52.9	28.0	45.0	47.3	39.9	17.3
PCA	41.2	60.3	58.8	52.9	58.8	35.6	55.5	51.3	36.7	28.4
PCA · corrected	57.4	54.4	50.0	54.4	58.8	50.0	48.7	39.6	48.5	28.2
Supervised	42.6	52.9	52.9	48.5	52.9	31.9	42.4	40.1	41.0	25.6
Supervised · corrected	44.1	52.9	51.5	51.5	52.9	30.8	41.5	39.8	40.6	25.6
Supervised + PCA	54.4	50.0	48.5	51.5	51.5	26.3	32.7	30.6	32.4	29.2
Supervised + PCA · corrected	45.6	50.0	47.1	51.5	51.5	32.5	38.8	34.4	37.4	29.8
Full tokens	N/A	N/A	45.6	N/A	N/A	N/A	N/A	18.8	N/A	N/A
Full tokens · corrected	N/A	N/A	45.6	N/A	N/A	N/A	N/A	25.7	N/A	N/A
CellFM 20-view										
Full	79.4	75.0	70.6	61.8	79.4	72.2	72.5	52.8	41.8	64.4
Full · corrected	73.5	75.0	66.2	48.5	79.4	67.3	72.5	57.3	46.9	64.4
PCA	79.4	72.1	79.4	75.0	79.4	63.1	70.2	77.7	70.9	66.0
PCA · corrected	83.8	72.1	75.0	77.9	80.9	68.3	68.9	72.2	74.3	71.6
Supervised	70.6	66.2	69.1	66.2	73.5	58.7	64.9	63.3	62.8	66.5
Supervised · corrected	77.9	66.2	72.1	67.6	73.5	53.9	63.4	64.3	64.4	66.5
Supervised + PCA	72.1	66.2	66.2	64.7	73.5	59.9	66.7	58.0	57.2	54.0
Supervised + PCA · corrected	72.1	66.2	69.1	66.2	69.1	52.7	63.3	64.8	63.7	57.9
Full tokens	N/A	N/A	79.4	N/A	N/A	N/A	N/A	64.9	N/A	N/A
Full tokens · corrected	N/A	N/A	73.5	N/A	N/A	N/A	N/A	47.3	N/A	N/A
BulkRNABert										
Full	72.1	61.8	64.7	72.1	76.5	68.8	61.3	56.3	61.5	68.0
Full · corrected	72.1	67.6	72.1	70.6	52.9	42.4	38.9	42.5	40.7	17.3
PCA	55.9	61.8	60.3	57.4	69.1	48.7	62.1	58.9	56.5	47.0
PCA · corrected	72.1	58.8	60.3	80.9	52.9	45.5	28.5	30.9	59.6	17.3
Supervised	69.1	58.8	63.2	69.1	75.0	68.4	58.4	52.3	61.7	62.6
Supervised · corrected	69.1	57.4	73.5	73.5	54.4	49.2	27.7	46.1	43.1	20.6
Supervised + PCA	57.4	54.4	60.3	60.3	75.0	40.5	56.9	59.5	61.1	52.9
Supervised + PCA · corrected	64.7	55.9	61.8	67.6	52.9	36.7	25.4	33.4	44.8	17.3

d

CLLmap — foundation-model performance

Classifiers

	Accuracy					Macro-F1				
	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM	Omiclass	Logistic Regression	MLP	MLP + SMOTE	SVM
DE baseline										
DE	95.2 ±2.4	N/A	N/A	N/A	N/A	95.2 ±2.4	N/A	N/A	N/A	N/A
DE · corrected	95.8 ±2.4	N/A	N/A	N/A	N/A	95.8 ±2.4	N/A	N/A	N/A	N/A
BulkFormer										
Full	66.7	73.6	79.3	44.8	81.6	66.6	73.5	79.0	39.9	81.3
Full · corrected	82.8	79.3	75.9	59.8	77.0	82.2	78.5	74.5	58.4	76.3
PCA	72.4	57.5	67.8	64.4	67.8	72.2	56.6	67.8	64.3	67.8
PCA · corrected	78.2	80.5	74.7	75.9	80.5	77.7	79.8	73.9	74.8	80.1
Supervised	69.0	79.3	77.0	74.7	74.7	68.8	79.0	74.6	74.4	73.4
Supervised · corrected	81.6	80.5	81.6	71.3	79.3	81.0	80.1	79.6	71.2	78.5
Supervised + PCA	72.4	79.3	78.2	78.2	73.6	72.4	78.5	77.2	77.4	73.1
Supervised + PCA · corrected	75.9	82.8	80.5	83.9	80.5	75.4	82.0	80.1	83.4	80.4
Full tokens	N/A	N/A	83.9	N/A	N/A	N/A	N/A	83.4	N/A	N/A
Full tokens · corrected	N/A	N/A	87.4	N/A	N/A	N/A	N/A	86.8	N/A	N/A
CellFM 1-view										
Full	52.9	60.9	55.2	66.7	56.3	50.2	56.0	55.1	63.4	47.9
Full · corrected	52.9	60.9	62.1	57.5	56.3	45.5	56.0	59.5	56.0	47.9
PCA	58.6	59.8	49.4	51.7	62.1	53.4	55.1	46.9	49.3	51.9
PCA · corrected	50.6	59.8	50.6	48.3	62.1	41.7	55.1	47.8	46.5	51.9
Supervised	49.4	50.6	54.0	54.0	49.4	40.9	43.9	51.2	51.7	42.1
Supervised · corrected	50.6	50.6	52.9	57.5	49.4	43.9	43.9	49.0	54.6	42.1
Supervised + PCA	51.7	52.9	51.7	54.0	54.0	44.7	46.5	48.1	49.9	49.9
Supervised + PCA · corrected	55.2	51.7	47.1	44.8	52.9	43.2	46.6	41.5	38.9	49.0
Full tokens	N/A	N/A	54.0	N/A	N/A	N/A	N/A	53.9	N/A	N/A
Full tokens · corrected	N/A	N/A	52.9	N/A	N/A	N/A	N/A	52.8	N/A	N/A
CellFM 20-view										
Full	74.7	81.6	77.0	66.7	83.9	74.3	81.2	77.0	66.5	83.4
Full · corrected	82.8	81.6	78.2	78.2	83.9	82.2	81.2	78.1	78.1	83.4
PCA	82.8	86.2	87.4	86.2	88.5	82.4	85.9	86.9	85.7	88.0
PCA · corrected	81.6	86.2	85.1	86.2	88.5	81.3	85.9	84.5	85.7	88.0
Supervised	74.7	80.5	79.3	71.3	77.0	74.4	79.8	78.7	71.2	76.5
Supervised · corrected	67.8	77.0	80.5	69.0	79.3	67.6	76.5	80.1	68.9	78.7
Supervised + PCA	66.7	77.0	75.9	73.6	73.6	66.6	76.6	75.7	73.5	72.9
Supervised + PCA · corrected	67.8	78.2	75.9	72.4	77.0	67.8	77.9	75.4	72.1	76.1
Full tokens	N/A	N/A	93.1	N/A	N/A	N/A	N/A	92.7	N/A	N/A
Full tokens · corrected	N/A	N/A	90.8	N/A	N/A	N/A	N/A	90.0	N/A	N/A
BulkRNABert										
Full	65.5	79.3	64.4	59.8	69.0	65.5	79.1	64.3	59.2	68.8
Full · corrected	79.3	82.8	86.2	80.5	77.0	77.4	81.6	85.8	80.2	74.9
PCA	65.5	70.1	78.2	57.5	74.7	65.5	70.1	78.0	56.3	74.7
PCA · corrected	77.0	69.0	77.0	78.2	69.0	75.3	68.9	76.9	78.1	68.8
Supervised	72.4	72.4	54.0	59.8	65.5	72.3	72.3	52.2	59.2	65.4
Supervised · corrected	78.2	81.6	72.4	85.1	75.9	75.6	80.2	72.4	84.7	74.2
Supervised + PCA	64.4	75.9	69.0	69.0	69.0	64.3	75.8	69.0	68.9	68.9
Supervised + PCA · corrected	75.9	75.9	81.6	73.6	67.8	74.8	75.4	81.3	73.5	67.6